

RESEARCH ARTICLE

Woody Debris Amendment Enhances Reclamation after Oil Sands Mining in Alberta, Canada

Robyn L. Brown¹ and M. Anne Naeth^{2,3}

Abstract

Mining disturbs large forested areas around the world, including boreal forests after oil sands mining in Canada. Industrial companies are expected to reclaim degraded land to ecosystems with equivalent land capability. This research showed the value of woody debris for reclamation of dramatically disturbed landscapes with a forest ecosystem end land use. Adding woody debris during reclamation can facilitate recovery of flora, soil nutrient cycling and water and nutrient holding capacity. Combined with forest floor material, woody debris can provide native plant propagules that would be otherwise commercially unavailable. Sites with and without woody debris on forest floor material containing identifiable litter (L), fragmented and fermented litter (F), and humus (H) (LFH), and peat mineral soil mix (peat) cover soils were studied. Within 2

years, woody debris decreased bare ground and created microsites which were positively associated with greater vegetation cover and woody plant density. Woody debris treatments had lower soil available nitrate and soil under woody debris had a lower temperature range and higher soil volumetric water content than control treatments without woody debris. Woody debris did not affect first year microbial biomass carbon or mycorrhizae, but both were greater on LFH than peat cover soil. LFH was associated with lower bare ground and greater vegetation cover, species richness, and soil phosphorus and potassium than peat cover soil, which had greater soil sulfate.

Key words: boreal forest, cover soil, LFH, microbial biomass carbon, mycorrhizae, nitrogen.

Introduction

Mining disturbs large areas of forest in numerous global ecosystems (Krankina et al. 1994; Zhang et al. 2011; Krümmelbein et al. 2012), including oil sands mining in Alberta, Canada which has garnered international attention. Large quantities of woody debris accumulate as lands are cleared for mining and are considered a waste product of mining. Woody debris is all dead woody material including above ground logs, limbs, twigs, and woody fruit; below ground buried wood and dead roots; and standing snags and stumps (Pyle & Brown 1999). Merchantable timber is sold and remaining woody material is burnt in piles or disposed of in large pits. Alternative uses for industrial waste products, including woody debris and forest floor materials, are of growing interest as disposal costs increase and availability of organic matter and nutrient sources for reclamation are sparse.

Few studies evaluated benefits of woody debris in reclamation, except to note it reduces soil erosion (Whisenant 2005). In natural forest ecosystems, woody debris creates microsites

that provide habitat and shelter for organisms, increases microbial presence, aids nutrient cycling, and increases plant growth and productivity (Harmon et al. 1986; Pyle & Brown 1999; Debeljak 2006). These positive effects of forest floor woody debris could be of considerable benefit on anthropogenic soils and reclamation sites.

Reclamation of surface mined lands relies on cover soils to restore vegetation diversity and soil function. Cover soils increase organic matter, nutrients, microorganisms, water holding capacity (Hargis & Redente 1984; Winter Sydnor & Redente 2002), and native plant seeds and propagules (MacKenzie & Naeth 2007, 2010). Common cover soils in boreal forests include upper soil horizon material mixed with peat or LFH (identifiable litter (L), fragmented and fermented litter (F), and humus (H)) (Alberta Environment 2007). Peat mineral soil mix (hereafter peat) consists of 1 m peat per 0.4 m underlying mineral soil (Oil Sands Vegetation Reclamation Committee 1998). LFH mineral soil mix (hereafter LFH) is a mix of LFH and Ae horizons from Luvisolic soils, including fine roots and tree stumps.

Reclamation requires ecosystem rebuilding and industrial companies are expected to reclaim land to equivalent capability relative to predisturbance conditions (Oil Sands Vegetation Reclamation Committee 1998). Innovative methods to reclaim soil and plant communities are required to jump start succession toward natural forest. As woody debris has ecological value in forest ecosystems it was hypothesized to have similar

¹Cardno ENTRIX, 801 Second Avenue, Suite 700, Seattle, WA 98104, U.S.A.

²Department of Renewable Resources, University of Alberta, Edmonton, Alberta T6G 2H1, Canada

³Address correspondence to M. A. Naeth, email anne.naeth@ualberta.ca

value in reclamation. This study aimed to determine if applying *Populus tremuloides* Michx. (trembling aspen) mixed wood and *Picea mariana* Mill. BSP. (black spruce) woody debris over peat or LFH cover soil would enhance reclamation by aiding recovery of native vegetation, soil nutrients, soil temperature and water regimes and microbial biomass.

Methods

Research Site

Research was conducted at Suncor Energy Inc., 24 km north of Fort McMurray, Canada, within the central mixed wood natural subregion of the boreal forest natural region (Natural Regions Committee 2006). Winters are long and cold, summers are short and warm. Mean annual temperatures are 0.7°C; mean maximum temperature occurs in July (23.2°C) and mean minimum temperature in January (−24°C). Average annual precipitation is 456 mm; 342 mm rain and 156 cm snow (Environment Canada 2003).

The central mixed wood natural subregion consists of large upland forest, wetlands, and rolling plains. Upland forests are composed of *Populus tremuloides* and *Picea glauca* (Moench) Voss (white spruce) mixed wood stands and Gray Luvisolic soil with some Dystric and Eutric Brunisols (Soil Classification Working Group 1998). Wetlands are dominated by *Picea mariana* bogs and fens, with *Salix* (willow) and sedge species. Bogs and fens have organic soils, predominantly Mesisols (Natural Regions Committee 2006). Soils in all treatments had loam textures, with peat pH averaging 6.5 and LFH 7.5 (Brown 2010). Electrical conductivity and sodium adsorption ratio were rated good according to Alberta Tier 1 guidelines (Alberta Environment 1994). Total organic carbon was higher in peat than LFH (7.39 vs. 5.52%) at the beginning and end (6.35 vs. 5.93%) of the study.

Experimental Design

Research plots were established on a slight east facing slope. The site was cleared in 1999 and used as a saline sodic overburden dump until 2004. Between November 2007 and February 2008, 36 plots were established to study LFH and peat cover soils with three woody debris treatments (no woody debris, woody debris from *P. mariana* forest, and woody debris from *P. tremuloides* mixed wood forest). Two rows of plots were arranged horizontally along the slope in a complete randomized block. Each plot was 10 m wide and 30 m long; each row contained three replicates of each treatment. Plots were separated by 5 m and rows by 10 m.

Half the plots received LFH and half received peat. LFH was salvaged to 20 cm and applied at 20 cm depth over 30 cm of mixed B and C horizon subsoil and 100 cm of clean overburden. Peat was applied at 30 cm depth over 100 cm of clean overburden. Covers were applied between November and December 2007 with a bulldozer. For each cover type, six plots were spread with *P. mariana* woody debris, six with *P. tremuloides* mixed wood woody debris, and

Table 1. Mean cover (%) of woody debris per size class (per 1 m²) shortly after plot establishment ($n = 6$).

	LFH		Peat	
	<i>Picea mariana</i>	<i>Populus tremuloides</i>	<i>Picea mariana</i>	<i>Populus tremuloides</i>
Fine	10.3 (1.2)	2.2 (0.4)	6.2 (1.9)	1.4 (0.1)
Small	6.9 (0.9)	2.9 (0.3)	4.7 (0.8)	2.2 (0.4)
Medium	13.1 (1.0)	8.1 (0.6)	13.7 (1.7)	9.1 (1.1)
Large	0.5 (0.4)	2.7 (1.0)	2.5 (0.5)	5.4 (1.4)
Total	30.9 (1.5)	15.9 (1.5)	27.1 (3.3)	18.1 (1.7)

Means are followed by SEs in brackets.

six controls had no woody debris. *Populus tremuloides* mixed woody debris contained approximately 70% *P. tremuloides*, 30% *Picea glauca*, and traces of *Betula papyrifera* Marsh. (paper birch). Woody debris was fresh salvaged and applied in February 2008 with a backhoe with grapple device. Debris was arranged to provide maximum contact with soil and spaced far enough apart for assessment in areas of woody debris and areas without woody debris. Woody debris was divided into four cover classes based on diameter: fine (≤ 2 cm), small (2–15 cm), medium (5–15 cm), and large (> 15 cm). Average woody debris cover was 15–30% of a plot at the start of the study (Table 1). *Picea mariana* provided considerably more cover than *P. tremuloides* due to a greater abundance of small and fine woody debris. Plots received aerial applications of N:P₂O₅:K₂O fertilizer as per standard reclamation practice; 23.5:25.0:8.0 at 300 kg/ha in June 2008 and 31.5:16.0:5.0 at 250 kg/ha in August 2009.

Sampling and assessment times were set for the beginning and end of two growing seasons. These two annual time frames allowed for determination of short-term changes in treatments.

Soil Sampling and Analyses

Soil was auger sampled in May and August 2008 and June and August 2009 at 0–10 cm depth. In each woody debris plot, three samples were collected and composited from under large and under small woody debris and in controls. Samples were sealed in plastic bags and refrigerated for three weeks at 4°C until analyzed at a commercial laboratory. Available ammonium was determined by potassium chloride extraction (Carter 1993); available nitrate, phosphorus, and potassium by the modified Kelowna method (Qian et al. 1994) and; available sulfate by calcium chloride extraction (McKeague 1978).

HOB0 MicroStation data loggers (Model H21-002; Onset Computer Corporation, Bourne, MA, U.S.A.) with plug in ECH₂O soil volumetric water smart sensors (Decagon Devices Inc., Pullman, WA, U.S.A) and 12-bit temperature smart sensors (Onset Computer Corporation) were installed in the bottom row of plots in May 2008. Data loggers were installed on level ground at 5 cm depths under large and small woody debris in each treatment and control. Data was collected hourly

Table 2. Mean available soil nutrients during the first (2008) and second (2009) growing seasons for cover treatments ($n = 18$).

		First Growing Season		Second Growing Season	
		LFH	Peat	LFH	Peat
Nitrate	mg/kg	4.5 (0.8)	6.1 (0.9)	2.0 (0.0)	2.2 (0.2)
Ammonium	mg/kg	1.0 (0.1) ^a	<0.3 ^b	—	—
Phosphorus	mg/kg	21.3 (1.3) ^a	7.1 (0.8) ^b	32.4 (2.2) ^a	16.0 (1.8) ^b
Potassium	mg/kg	—	—	137.0 (13.1) ^a	97.0 (6.8) ^b
Sulfate	mg/kg	—	—	31.3 (8.8) ^b	133.3 (14.0) ^a

Means are followed by SEs in brackets. Different letters denote significance between treatments within a given sampling date at $p \leq 0.100$.

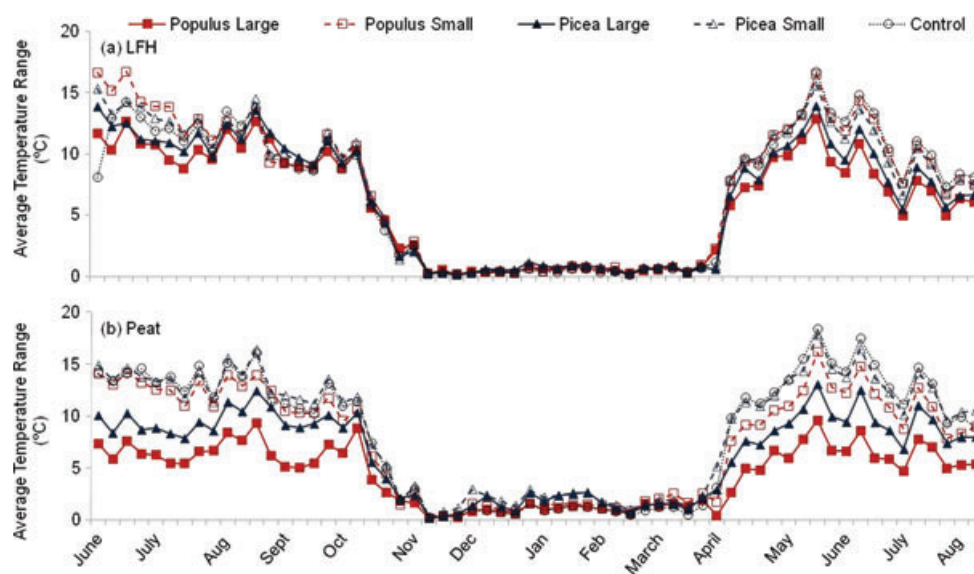


Figure 1. Mean weekly soil temperature range ($^{\circ}\text{C}$) at 5 cm depth, starting 1 June 2008 and ending 8 August 2009 for (a) treatments with LFH cover soil and (b) treatments with peat cover soil.

and mean weekly volumetric water content and temperature ranges calculated per treatment location. A calibration equation was used for volumetric water content for each cover type.

Microbial Sampling and Analyses

In August 2008, soil samples for microbial biomass carbon were collected from 0–10 cm depths. In each treatment, two samples were collected and composited under each of large woody debris and small woody debris and in controls. Samples were double bagged, placed on dry ice (-78°C) and stored in a freezer (-20°C) until analyzed by chloroform fumigation extraction (Vance et al. 1987) in December 2008.

To determine mycorrhizal biomass, roots from *Achillea millefolium* L. (common yarrow), *Geranium bicknellii* Britt. (northern cranesbill), and *Rubus idaeus* L. (wild red raspberry) were collected from each plot. Only plants next to woody debris were collected from woody debris plots. Roots were extracted from soil, double bagged, placed on dry ice (-78°C) and stored in a freezer (-20°C) until analysis between November 2008 and February 2009. Glucosamine ($\text{C}_6\text{H}_{13}\text{NO}_5$), an amino sugar in fungal cell walls, was analyzed to determine root mycorrhizal biomass by modified

procedures (Nilsson & Bjurman 1998; Appuhn et al. 2004; Appuhn & Joergensen 2006).

Vegetation Assessments

Detailed vegetation assessments were conducted in July 2008 and June and August 2009. Three permanent line transects with five evenly spaced 1 m^2 quadrats were established in each plot without moving woody debris. Cover was estimated in each quadrat for plant species, moss, woody debris, and bare ground. Woody plant densities were determined in July and August 2008 and June and August 2009. Woody debris overlapped by vegetation was included, thus cover could be greater than 100%. Covers less than 0.5% were recorded as trace. Only vegetation rooted inside quadrats and *Fragaria* (strawberry) runners were included. Species richness per quadrat was determined. Species nomenclature followed Moss (1994).

To determine if woody debris created microsites, proximity of plants to woody debris was assessed on peat with woody debris in June and August 2009. Woody plant density and plant cover were classified as near or away from woody debris. LFH plots were not assessed as vegetation was too dense and

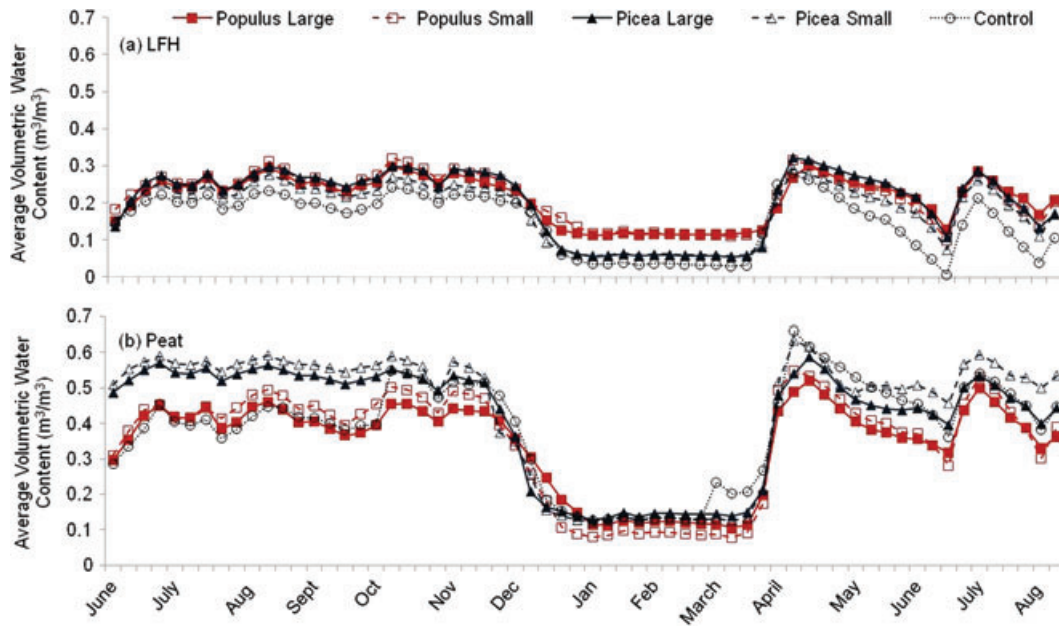


Figure 2. Mean weekly soil water content (m^3/m^3) at 5 cm depth, starting 1 June 2008 and ending 8 August 2009 for (a) treatments with LFH cover soil and (b) treatments with peat cover soil.

microsites created by vegetation could influence results. Plants within 10 cm of large or medium size logs or within diameter lengths of small woody debris were considered growing in a woody debris microsite, or near woody debris.

Statistical Analyses

Fresh woody debris cannot be utilized as substrate for plants, therefore the proportion of ground for rooting was calculated as: available ground = $100 - (\% \text{ woody debris cover} + \% \text{ rock cover})$. Plant species were classified by morphology, origin, and life history. Unknown plants were included in morphology and total groupings for cover. Vegetation not identified to species (*Carex*, *Salix*, and moss) was excluded from plant origin and species richness analyses. Trace values were set to 0.1%. Woody root material from LFH was excluded in woody debris cover assessments, but included for regression analyses.

PASW/SPSS 17 statistical software was used for analyses. Shapiro-Wilk's test was used for normality and Levene's test for equality of variance. Most data failed homogeneity of variance and normal distribution assumptions. Therefore, a Scheirer Ray Hare extension of the Kruskal–Wallis test was used as a nonparametric equivalent to two way analysis of variance (ANOVA) (Sokal & Rohlf 1995). As this test is conservative (Dytham 2003), significance was determined at $p \leq 0.100$ to increase power (Zar 1999). Significant differences were determined with a Mann–Whitney rank sum test.

Three-way ANOVA was used to analyze glucosamine data parametrically as a nonparametric equivalent does not exist. Data passed normality, but failed homogeneity of variance at $p = 0.031$. Parametric ANOVA is robust and variance is assumed sufficient to make this the preferred analysis.

Vegetation per available ground and woody plant density were regressed against woody debris cover. *T*-tests were used to analyze vegetation proportions near and away from woody debris. Parametric and nonparametric tests were conducted with the same results, therefore only parametric *p*-values are shown. Significance level for parametric ANOVA, regression, and *t*-tests was $p \leq 0.050$.

Results

No significant interactions occurred between cover soils and woody debris treatments for soil or vegetation response; therefore results of main effects are presented. Results are presented for the end of 2008 and 2009 growing seasons unless significant differences occurred between spring and late summer sampling times within a year.

Effects of Cover Materials and Woody Debris on Soil Properties

Cover type had significant effects on plant available nutrients. Phosphorus, potassium, and ammonium were highest with LFH; sulfate was highest with peat (Table 2). Temperature range was slightly lower (Fig. 1) and volumetric water content was higher (Fig. 2) with peat. Microbial biomass carbon was greater with LFH (mean 47.1 ± 3.8 SE ppm) than peat (11.6 ± 1.8 ppm) ($p < 0.001$). LFH (47.1 ± 3.8 ppm) had significantly more root glucosamine than peat (1.4 ± 0.1 mg/g) ($p = 0.028$).

Woody debris impacted soil available nitrate, temperature, and water content. In the first growing season, nitrate was significantly greater without woody debris (9.3 ± 1.9 mg/kg) than under large *Populus tremuloides* (3.7 ± 0.9 mg/kg) ($p = 0.024$).

Table 3. Mean bare ground, vegetation cover, and woody species density in the first (2008) and second (2009) growing seasons for cover ($n = 18$) and woody debris treatments ($n = 12$).

	Cover		Woody debris		
	LFH	Peat	<i>Populus tremuloides</i>	<i>Picea mariana</i>	Control
Bare ground (%)					
First growing season	66.9 (3.3) ^B	80.3 (2.9) ^A	73.4 (3.0) ^b	61.7 (3.4) ^c	86.1 (3.3) ^a
Second growing season	16.0 (1.7) ^B	50.2 (4.5) ^A	32.3 (5.25)	28.2 (6.1)	40.0 (5.9)
Vegetation per available ground (%)					
First growing season	12.8 (2.9) ^A	2.3 (1.2) ^B	6.2 (2.9)	8.7 (3.6)	7.7 (2.9)
Second growing season	65.1 (2.2) ^A	33.9 (1.8) ^B	49.5 (4.4)	52.6 (6.8)	46.3 (4.2)
Woody species density (plants m ⁻²)					
First growing season	24 (3) ^A	13 (2) ^B	13 (2) ^b	29 (3) ^a	14 (4) ^b
Second growing season	39 (7)	32 (12)	29 (7) ^b	52 (4) ^a	26 (6) ^b

Means are followed by SEs in brackets. Different letters denote significant differences within cover (upper case) and woody debris (lower case) treatments within a sampling date at $p \leq 0.100$; except woody species density $p \leq 0.050$.

Table 4. Mean cover (%) per species group in the first (2008) and second (2009) growing seasons for cover treatments ($n = 18$).

Plant group	First Growing Season		Second Growing Season	
	LFH	Peat	LFH	Peat
Forb	9.8 (2.4) ^a	2.0 (1.2) ^b	41.7 (2.7) ^a	26.6 (2.9) ^b
Pteridophyte	<0.1 ^b	0.1 (0.0) ^a	0.1 (0.1) ^b	1.3 (0.5) ^a
Grass	0.4 (0.1) ^a	0.1 (0.1) ^b	8.6 (2.2) ^a	1.5 (0.8) ^b
Sedge	0.1 (0.0) ^a	<0.1 ^b	6.2 (1.7) ^a	0.6 (0.2) ^b
Woody	0.2 (0.0) ^a	<0.1 ^b	3.9 (1.0) ^a	1.8 (0.6) ^b
Moss	0.25 (0.1)	0.5 (0.1)	16.5 (4.5) ^a	5.7 (1.8) ^b
Perennial	3.8 (0.9) ^a	0.6 (0.2) ^b	39.1 (5.05) ^a	15.1 (3.0) ^b
Annual / Biennial	6.8 (1.6) ^a	1.7 (1.1) ^b	21.5 (3.25) ^a	16.7 (2.3) ^b
Native	6.0 (1.4) ^a	1.1 (0.6) ^b	43.4 (4.1) ^a	15.3 (2.5) ^b
Non-native	4.4 (1.1) ^a	1.1 (0.7) ^b	10.4 (1.45) ^b	15.2 (2.5) ^a
Total With Moss	10.9 (2.5) ^a	2.8 (1.3) ^b	77.2 (4.1) ^a	37.6 (4.0) ^b

Means are followed by SEs in brackets. Different letters denote significance between treatments within a sampling date at $p \leq 0.100$.

and *Picea mariana* (3.3 ± 0.8 mg/kg) ($p = 0.011$). Nitrate was numerically higher in controls than under small woody debris. Trends continued during year 2, but woody debris treatment values were under detection limits (data not shown).

Weekly soil temperature range was generally lowest under large woody debris (Fig. 1), approximately 2°C lower in year 1 with LFH and 3°C lower in year 2 under large logs of either species than in controls. A similar trend occurred with peat where temperature range was 7°C lower under large *P. tremuloides* and 5°C lower under large *P. mariana* than controls (Fig. 1). With LFH, weekly water content was consistently lowest in controls, with no trend for woody debris species or size (Fig. 2). During snowmelt 2009, peat without woody debris peaked earlier and water content was greater than with woody debris (Fig. 2). Water content decreased in winter as sensors record free water in soil pores, not ice.

Effects of Cover Materials and Woody Debris on Vegetation

Cover type had significant effects on vegetation. LFH consistently had significantly less bare ground and greater cover per

available ground than peat; woody density was significantly greater with LFH the first growing season (Table 3). LFH had significantly greater cover of most plant groups except pteridophytes in both growing seasons (Table 4). Non-native species cover was significantly greater with LFH the first growing season, but significantly less the second and peat had a greater proportion of non-native to native species. Species richness was significantly greater with LFH than with peat in first (LFH = 20 ± 2 , Peat = 12 ± 1) ($p = 0.002$) and second growing seasons (LFH = 34 ± 1 , Peat = 25 ± 1) ($p < 0.001$).

Bare ground decreased with woody debris, significantly so with *P. tremuloides* ($p = 0.018$) and *P. mariana* ($p < 0.001$) in the first season (Table 3). Regression of vegetation cover per available ground versus woody debris cover was significantly positive with LFH throughout year 2 ($r^2 = 0.413$, $p = 0.004$; $r^2 = 0.405$, $p = 0.005$), but only with peat in June of year 2 ($r^2 = 0.225$, $p = 0.047$) (Fig. 3).

Few significant differences in canopy cover or species richness (data not shown) occurred in response to woody debris. Moss cover was significantly greater in *P. mariana*

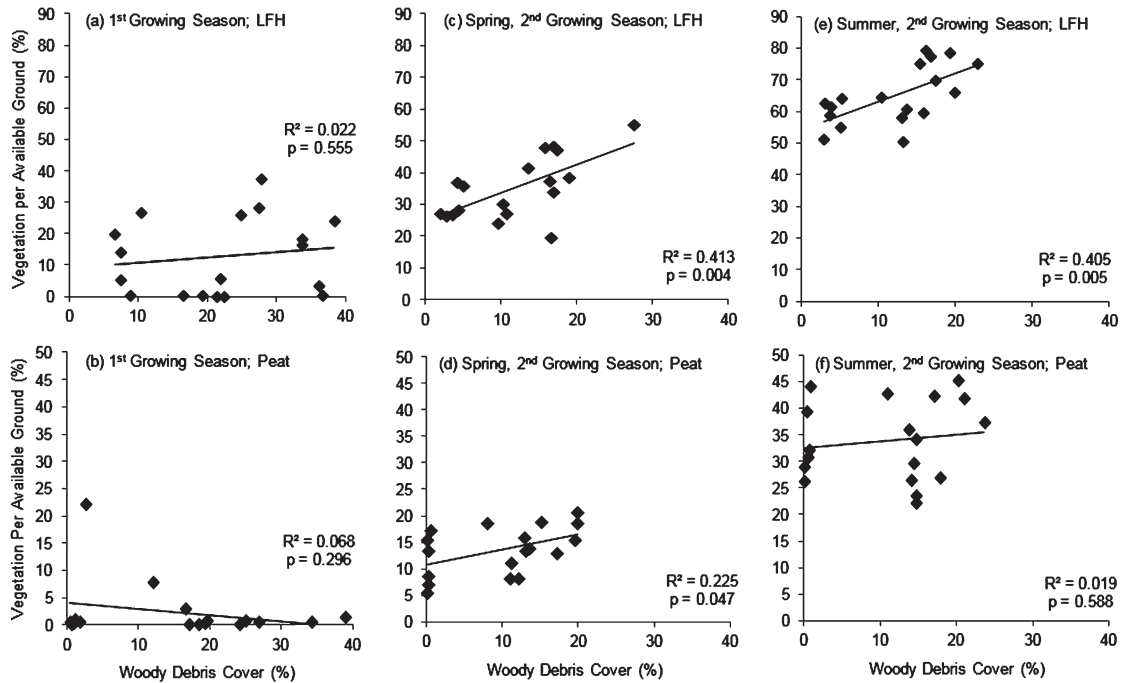


Figure 3. Regression analysis for vegetation canopy cover per available ground and woody debris ($n = 18$) during (a) July of first (2008) growing season on LFH and (b) peat cover soils, during the (c) spring (June) of the second (2009) growing season on LFH and (d) peat cover soils, and during the (e) summer (August) of the second growing season on LFH and (f) peat cover soils.

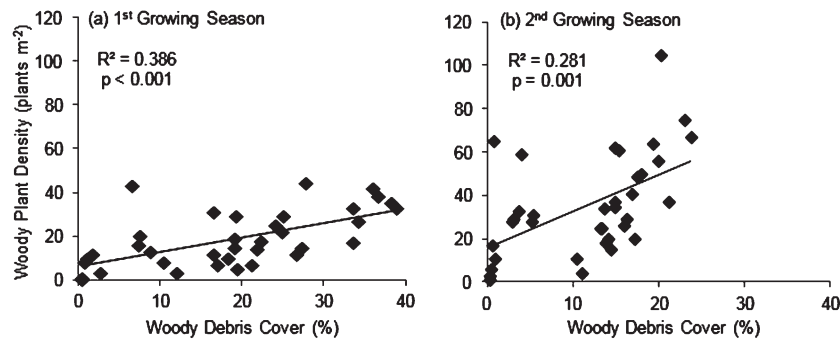


Figure 4. Regression analysis for woody plant abundance per plot and woody debris cover ($n = 36$) during (a) August of the first (2008) and (b) second (2009) growing seasons for both LFH and peat cover soils.

($0.7 \pm 0.2\%$) than controls ($0.1 \pm 0.0\%$) the first growing season ($p = 0.015$). Total plant cover was low for the first growing season on all woody debris treatments (5.4–8.15%) increasing considerably by year 2 (55–59%). During year 1, five of six peat control plots had vegetation cover less than 1% and one had greater than 21% due to the non-native annual *Chenopodium album* L. (white goosefoot).

Woody plant density was significantly different among woody debris treatments in first ($p = 0.002$) and second ($p = 0.001$) growing seasons (Table 3). *Picea mariana* treatments had significantly greater woody plant density throughout the study (Table 3). *Picea mariana* debris contained cones that produced viable seedlings; this treatment still had highest woody plant density not including *P. mariana* seedlings. A significant positive relationship was found

between woody plant density and woody debris cover in year 1 ($r^2 = 0.386$, $p < 0.001$) and year 2 ($r^2 = 0.281$, $p = 0.001$) (Fig. 4).

There was significantly more vegetation near woody debris in June but not August of the second growing season (Table 5). In June, cover of vascular plants ($p < 0.001$), moss ($p = 0.001$), total vegetation ($p < 0.001$), and non-native species ($p < 0.001$) was significantly greater near woody debris. More native species grew near woody debris. By August only moss cover was significantly greater near woody debris ($p < 0.001$) and non-native species cover was significantly greater away from woody debris ($p = 0.002$). Woody plant cover and density were significantly greater near woody debris in June ($p < 0.001$) and August ($p < 0.001$). *Salix* species were the most abundant woody plants and significantly

Table 5. Mean proportion of vegetation (%) located near and away from woody debris in peat treatments during the second growing season for microsite analysis ($n = 12$). Woody debris treatments are pooled.

	June 2009		August 2009	
	Near ¹	Away	Near	Away
Cover				
Total	59 (3) ^a	41 (3) ^b	50 (2)	50 (2)
Vascular plant	60 (3) ^a	40 (3) ^b	49 (3)	51 (3)
Moss	68 (7) ^a	32 (7) ^b	63 (4) ^a	37 (4) ^b
Native	53 (4)	47 (4)	51 (4)	49 (4)
Non-native	65 (4) ^a	35 (4) ^b	45 (2) ^b	55 (2) ^a
Woody species				
Density	77 (4) ^a	23 (4) ^b	71 (5) ^a	29 (5) ^b
Cover	70 (6) ^a	30 (6) ^b	69 (6) ^a	31 (6) ^b
<i>Salix</i> spp. density	84 (5) ^a	16 (5) ^b	87 (4) ^a	13 (4) ^b

Numbers are means followed by SEs in brackets. Different letters denote significance between near and away treatments within a sampling date at $p \leq 0.050$.

¹Near was defined as within 10 cm of woody debris and away as greater than 10 cm from woody debris.

denser near woody debris ($p < 0.001$). Over 80% of *Salix* were found near woody debris.

Discussion

This research shows the value of woody debris and LFH in reclamation. Microsites required for seed germination and seedling protection in harsh environments are provided by woody debris. Early successional sites often have extreme fluctuations in soil temperature and intense radiation, increasing evaporation and drying soil. Protection against water loss and seed desiccation is important for seed germination and plant establishment (Sheldon 1974; Jumpponen et al. 1999). Areas near and under large woody debris had muted temperature extremes and elevated soil water, providing microsites for seed germination and early seedling development.

Higher available nitrate in controls was likely associated with fertilization and possible nitrogen immobilization in woody debris treatments. Low nitrate can reduce early plant growth. Fertilizer would accumulate less under logs due to vertical obstruction. Several studies found nitrogen immobilization from saprotrophic fungi that translocated nutrients from soil into woody debris for decomposition (Wells & Boddy 1990). Forest soils with woody debris often have low soil nitrogen (Zimmerman et al. 1995; Kayahara et al. 1996), increasing in woody debris over time due to immobilization (Harmon et al. 1986; Krankina et al. 1999; Laiho & Prescott 1999). Substrate and environment strongly influence woody material decay (Laiho & Prescott 2004) with woody debris water content positively correlated with decomposition (Brown et al. 1995). In year 2, fungi were observed growing on logs in our study. Fungi are main decomposers of woody material and are expected to increase on woody debris, increasing the fungal bacterial ratio with time (Brant et al. 2006) and influencing nutrient cycling.

Woody debris, particularly with LFH, positively influenced successional trajectories, supporting previous work showing LFH is better cover soil than peat by increasing canopy cover, species richness, and diversity (MacKenzie & Naeth 2010). Woody debris provides further benefit by increasing woody plant density, a functional group, which is often difficult to establish on severely disturbed sites. During year 1, annual and biennial species dominated, due to quick emergence of *Potentilla norvegica* L. (rough cinquefoil), *Corydalis aurea* Willd. (golden corydalis), and *Geranium bicknellii*. By the second growing season, LFH had mostly perennial species; peat had mostly annuals and biennials and more non-native than native species. In year 1, main non-native species were *Chenopodium album* and *Crepis tectorum* L. (narrow leaf hawksbeard) which spread rapidly on peat in year 2. Wind dispersed seed of *Crepis tectorum* accumulated along woody debris. *Chenopodium*, an annual without wind dispersed seeds, was less affected by woody debris, explaining why non-native species cover was greater away from woody debris by the end of year 2. Moss grows best under vegetation, with low temperatures and radiation (Maestre et al. 2002) as seen with LFH in year 2. Non-native species were less of a problem on LFH due to greater cover of perennial native species in year 2. When LFH and woody debris were combined, more native plants were present and native species had a competitive advantage over non-native species.

According to regeneration niche theory, plant species have different requirements for survival and increasing habitat diversity should increase species richness (Grubb 1977). Reclaimed mine landscapes are generally homogeneous with few microsites to protect seedlings from climatic and biotic stressors. Adding woody debris increases landscape heterogeneity and should increase species richness. Microsites can provide greater opportunity for cohabitation (Harper et al. 1961) and increase evenness by not allowing one species to dominate. Woody debris attracted birds, which act as seed dispersal agents that could further increase diversity. Reclamation practices that require native species are often restricted by amount and kind of native species commercially available. If commercially unavailable seeds and propagules can be supplied by cover soils with large seed banks such as LFH, protection provided by woody debris could cause reclamation to proceed more rapidly with higher biodiversity.

Mine reclamation sites are often open with limited ground cover, exposing seedlings to wind and temperature extremes. Microsites with woody debris ameliorate conditions, providing favorable habitat, even in forest canopy gaps (Beach & Halpern 2001). Woody debris had a greater effect on woody plant density than cover soil type. Increased establishment with *Picea mariana* debris may reflect the greater variability in woody debris size classes compared to *Populus tremuloides*. *Picea mariana* woody debris acted as a propagule source for cones, which produced viable seedlings. *Picea mariana* are difficult to grow in oil sands reclaimed areas. Seedlings from cones on site may have better survival as seeds are selected for harsh early succession conditions. Woody species establishment on reclamation sites is associated with development

of a soil organic layer (Rowland et al. 2009), which aids ecological restoration, contributing to the value of woody debris in reclamation. Woody debris was positively associated with vegetation cover throughout year 2 with LFH and in spring with peat, indicative of succession processes favored by LFH with lower cover of non-native species than peat. Woody debris microsites likely enabled woody plants to survive harsh site conditions as seen by the positive association between woody plant density and woody debris.

This research shows the reclamation value of woody debris for severely disturbed landscapes with a forest ecosystem end land use. Benefits of woody debris microsites need to be balanced with potential negative effects of nutrient immobilization. Further research is needed to determine whether additional types of woody debris can be used, appropriate cover and volume of woody debris, whether covers that mimic mature undisturbed forests produce improved results, long-term successional patterns, and best practices for salvaging and incorporating woody debris.

Implications for Practice

- Woody debris can be used for reclamation of disturbed lands with a forest end land use.
- Woody debris can increase woody plant density by providing microsites with favorable growing conditions and providing cones as a source of locally adapted seed.
- Cover soils with large seed banks (such as LFH) function well with woody debris and produce favorable successional trajectories that are less dependent on commercially produced seed.
- Woody debris can immobilize nitrogen, which should be monitored on reclamation sites.

Acknowledgments

Funding for this research from Suncor Energy Inc., Syncrude Canada Ltd., Shell Albian Sands, and Total E&P Canada is gratefully acknowledged. Authors have no conflicts of interest.

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